Statistical packages usually offer two options for estimating variance. Let $x_1, \ldots, x_k$ be i.i.d. samples with mean $\mu$ and variance $\sigma^2$. The two estimators are

$$\hat{\sigma}_k^2 = \frac{1}{k-1}\sum_{i=1}^{k}(x_i - \hat{x})^2 \tag{1}$$

$$\hat{\sigma}_k^2 = \frac{1}{k}\sum_{i=1}^{k}(x_i - \hat{x})^2 \tag{2}$$

where $\hat{x} = \frac{1}{k}\sum_{i=1}^{k} x_i$ is the sample mean.

i) State the definition of the bias of an estimator.

ii) Choose *one* of the estimators (1) or (2) and derive its bias. Is the estimator biased? *Hint:* $\mathbf{E}[Z^2] = \mathrm{Var}(Z) + \mathbf{E}[Z]^2$.